

Understanding solid-state battery electrolytes using atomistic modelling and machine learning

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Abstract

Solid-state batteries that use solid electrolytes are attracting interest for their potential safety, stability and high energy density, making them ideal for next-generation technologies including electric vehicles and grid-scale renewable energy storage. Advances in solid electrolytes require the design and optimization of current and new materials, informed by a deeper understanding of their properties on the atomic and nanoscale. This Review highlights progress in using atomistic modelling and machine learning techniques to gain valuable insights into inorganic crystalline solid electrolytes for lithium-based and sodium-based batteries. We discuss computational studies on oxide, sulfide and halide materials that examine three fundamental properties critical to their performance as solid electrolytes: fast-ion conduction mechanisms, interfacial effects and chemical stability. The resulting insights help to identify design strategies for the future development of improved solid-state batteries.

Sections

Introduction

Fundamentals and challenges of solid-state batteries

Inorganic solid electrolyte materials

Materials modelling and machine learning methods

Understanding the mechanisms of fast-ion transport

Predicting phase formation, stability and decomposition

Understanding interfacial resistance and reactivity

Future perspectives

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Introduction

Energy storage has evolved into a priority area within global research, with governments and industry investing heavily to achieve net zero targets. The lithium-ion battery is essential to this challenging task and has already revolutionized the use of portable electronics worldwide. Nevertheless, major improvements are required to meet the current and future demands of electric vehicles and grid-scale renewable energy storage. The next generation of Li-ion batteries must be cheaper, safer, and higher in energy density. To achieve such ambitious advances, it is necessary to look beyond conventional Li-ion batteries for transformative change^{1–3}.

The solid-state battery, which uses a solid electrolyte rather than the flammable liquid electrolytes found in commercial Li-ion batteries, has the potential to improve the safety and energy density of Li-ion batteries^{4–6}. Increased safety results from isolating the electrodes with the solid electrolyte, avoiding Li build-up that can trigger short-circuiting. Energy density gains are possible with the use of an energy-dense Li-metal anode. In addition to lithium solid-state batteries, sodium solid-state batteries are also receiving notable attention for safe and economical energy storage⁷.

However, many fundamental challenges must be overcome to truly realize the exciting benefits of solid-state batteries. These challenges include insufficient ionic conductivity and stability, as well as numerous interfacial issues^{8,9}, which greatly depend on the properties of the constituent battery materials, particularly the solid electrolyte. Major innovation in solid electrolytes hinges on the design and optimization of current and new materials, which in turn require a deeper understanding of their atomic and nanoscale characteristics. Advanced computational techniques are now readily used to probe and elucidate the atomic-scale properties of functional materials, including mechanistic features that are often very challenging to characterize

with conventional experimental methods, such as ion conduction pathways and complex correlated mechanisms^{10–13}. With the rise of supercomputing power and the need to analyse enormous datasets for materials discovery, the opportunities for computational materials science have never been greater.

In this Review, we describe important contributions to lithium-based and sodium-based crystalline solid electrolytes for solid-state batteries that have been achieved through atomistic modelling and the increasing use of machine learning (ML) techniques. Reviews focusing on the experimental characterization, practical device aspects, and mesoscale and continuum-scale modelling of solid electrolytes, including polymers and hybrids, and solid-state batteries are available elsewhere^{4–9,14–21}. We begin by providing an overview of the solid-state battery concept, its challenges, and the families of inorganic crystalline solid electrolyte materials. Next, we briefly outline the methodologies behind atomistic simulation and ML techniques. We then turn our attention to the progress made through computational studies in three critically important aspects of solid electrolytes: fast-ion conduction mechanisms, chemical stability and interfacial effects.

Fundamentals and challenges of solid-state batteries

A solid-state battery, as schematically displayed in Fig. 1, has three main components: the metal anode (represented by a thin flat surface), the solid electrolyte and the cathode (the latter of which are both made up of polycrystalline particles). Cathode and solid electrolyte particles are mixed to form composite cathodes in solid-state batteries that help to establish efficient electron and ion conduction pathways. In addition to the safety benefits of using a solid electrolyte in place of a combustible liquid electrolyte, solid-state batteries offer the potential advantage of using Li or Na metal anodes and high-voltage (>5 V) cathode materials to design high-energy-density batteries. Inorganic solid electrolytes can also support battery operation at a wide temperature range (for example, –50 °C to 200 °C or more) in which conventional liquid electrolytes would freeze, boil or decompose. An additional advantage enabled by the use of a solid electrolyte is bipolar stacking of the anode of one cell and the cathode of the next cell on the same current collector to produce individual cells with higher voltage^{4,5}.

Despite extensive efforts to realize the advantages of solid-state batteries, they have yet to achieve their full potential. Several major challenges remain, primarily centred around the performance of the solid electrolyte and its interfaces (Fig. 1). It was initially believed that, owing to the mechanical rigidity of solid electrolytes, their use with metal anodes would prevent the growth of dendrites, which propagate from the anode to the cathode through the separator and cause short-circuit. However, it is now known that this is typically not the case, and that dendrites are still capable of propagating through solid electrolytes^{4,22}. Crack propagation at the solid electrolyte surface resulting from lithium or sodium metal being forced through microcracks at high pressures is another issue associated with the anode–electrolyte interface and is strongly linked with dendrite propagation. Interfacial stabilization and resistivity are key challenges at both electrolyte–electrode interfaces given that the interfacial composition and structure between solid electrolytes and electrode materials often deviate substantially from those of the bulk materials. Chemo-mechanical challenges, such as the maintenance of physical contact between solid particles during electrochemical cycling and the formation of a stable and conductive interphase, are important

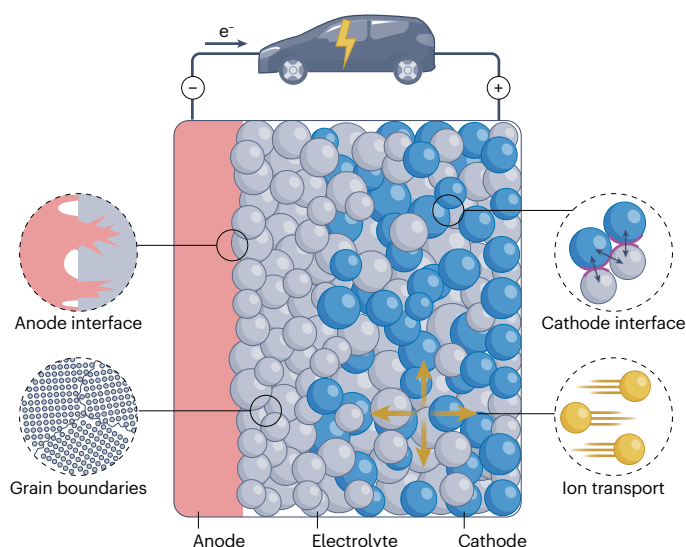


Fig. 1 | Components, interfaces and challenges of solid-state batteries.

Schematic of a solid-state battery highlighting its main components and some of the major challenges facing this technology. These challenges include dendrite growth resulting from the use of metal anodes, reduced stability, increased resistivity, cracking and porosity at the electrolyte–electrode interfaces, the maintenance of physical contact during electrochemical cycling between solid particles, and achieving sufficient ionic conductivity in the bulk materials and at the nanostructural and microstructural scales.

to maintaining ion diffusion between the electrodes and electrolyte. In fact, the use of a small fraction of liquid additive – that is, forming a hybrid solid-state battery – has been proposed to alleviate these issues and maintain long-term stability under the influence of cyclic volume changes of the active materials^{23–25}. Furthermore, the internal nanometre-scale to micrometre-scale interfaces that exist within crystalline solid electrolytes can cause issues because they tend to dominate the ionic conductivity. In particular, grain boundaries are known to act as defect sinks and can severely inhibit ionic conductivity in many solid electrolytes, especially oxide-based materials. In addition to the plethora of interfacial challenges, ensuring that the bulk solid electrolyte material has practical room-temperature ion conductivity ($>10^{-3}$ S cm⁻¹) remains fundamental to the use of solid-state batteries at ambient conditions^{4,5,8,9}.

Inorganic solid electrolyte materials

An ever-increasing array of crystalline ion conductors is being considered as candidate solid electrolytes for solid-state batteries. The crystal structures of some of the most topical lithium-based and sodium-based solid electrolyte materials are highlighted in Fig. 2a, with their corresponding ionic conductivities (σ) presented in Fig. 2b. These materials, each with their own particular strengths and weaknesses, can be broadly divided into subcategories of oxides, sulfides and halides.

Oxides

Of the Li-based oxides, garnet-structured materials, typified by Li₇La₃Zr₂O₁₂ (LLZO), have received particular interest owing to their high room-temperature conductivity (10^{-3} – 10^{-4} S cm⁻¹), chemical stability against Li metal and wide electrochemical potential window^{26–28}. However, the garnet family generally suffers from high grain-boundary resistance²⁹ and moisture sensitivity³⁰. Li-ion conducting perovskites (such as La_{0.66}Li_{0.33}TiO₃ (LLTO)) and antiperovskites (such as Li₂OHCl) also feature heavily in the literature^{31–34}. Although LLTO has reasonable room-temperature conductivity, negligible bulk electronic conductivity and a wide electrochemical window, its practical application has been severely hindered by its extremely low conductivity at its grain boundaries and by stability problems associated with its direct contact with Li metal³⁵. Antiperovskite solid electrolytes have wide electrochemical stability windows, soft mechanical properties and stability against Li metal but inferior conductivity compared with other oxide solid electrolytes³⁶.

Research into Na-ion conducting oxides has mostly centred on the β -alumina and Na superionic conductor (NASICON) structures, primarily owing to their excellent ionic conductivities. β -alumina is a layered material with alternating spinel blocks and conduction planes of mobile Na, Li, Ag or K ions. It has two main crystal structures, β -Al₂O₃ and β'' -Al₂O₃, with the latter generally exhibiting higher ionic conductivity ($>10^{-2}$ S cm⁻¹ at 300 °C for single crystals)^{37,38}. These materials, however, suffer from synthesis challenges related to difficulties in accurately controlling composition and phase content, poor interfacial stability and microstructural issues associated with abnormal grain growth and grain boundaries³⁹. NASICON electrolytes take the general formula of Na_{1+x}Zr₂Si_xP_{3-x}O₁₂ ($0 \leq x \leq 3$). In contrast to β -alumina, NASICONs are 3D ion conductors and typically exhibit Na-ion conductivities of 10^{-4} S cm⁻¹ at ambient temperature^{40,41}. They do, however, share some of the weaknesses of β -alumina, such as the formation of inhomogeneous products during their solid-state synthesis, as well as instability when in contact with Na metal⁴².

Sulfides

Sulfide electrolytes generally benefit from both higher ductility and ionic conductivity and reduced grain-boundary resistance compared with oxides. As a result, excellent contact with electrodes can be realized for sulfides by simple cold pressing, thereby making the fabrication of bulk solid-state batteries easier^{43,44}. The 2011 discovery of Li₁₀GeP₂S₁₂ (LGPS) with its remarkable Li-ion conductivity of 10^{-2} S cm⁻¹, comparable to or greater than that of commercial liquid electrolytes, was pivotal to the current surge of interest in solid-state batteries⁴⁵. Despite this considerable advantage, LGPS possesses a limited electrochemical window and poor interfacial stability with Li metal^{46,47}. The argyrodites, based on halogen-substituted Li₂PS₆ (Li₂PS₅X; X = Cl, Br or I), exhibit good ionic conductivity of 10^{-3} S cm⁻¹ (refs. 48–51). Nevertheless, Li argyrodites are highly sensitive to air and moisture and are also unstable against Li metal⁵². Other state-of-the-art sulfide electrolytes include a variety of materials, such as the Li–P–S system (including Li₃PS₄ (ref. 53) and Li₇P₃S₁₁ (ref. 54)), Li_{9.54}Si_{1.74}P_{1.44}S_{11.7}Cl_{0.3} (ref. 55) and Li₇Si₂S₇I (refs. 56,57), which have Li-ion conductivities comparable to those of liquid electrolytes.

Like their Li-ion counterparts, Na-based thiophosphate materials, exemplified by Na₃PS₄, dominate the research into sodium sulfide electrolytes owing to their promising ionic conductivity and ductility^{58,59}. For Na₃PS₄, most research has focused on cation (such as Sn, Sb, W and Ca)^{60,61} and anion (such as O, Se and Cl)⁶² substitutions to increase the room-temperature conductivity towards and beyond 10^{-2} S cm⁻¹. Additional Na-ion conducting thiophosphates include Na variations of LGPS, such as Na₁₁Sn₂PS₁₂ (ref. 63).

Halides

Halide solid electrolytes have attracted interest since the 2018 report⁶⁴ of superionic conductivity for Li₃YCl₆ and Li₃YBr₆. The Li₃MX₆ stoichiometry, where M is a trivalent cation and X is a halide, has been indexed to several structure types depending on the ratio of the ionic radii and the synthesis conditions. Specifically, face-centred cubic (fcc) chlorides with M = In (refs. 65,66) or Sc (refs. 67,68) possess both good ionic conductivity and a wide electrochemical window, with the In system also presenting excellent moisture resistance. A mixed In–Sc halide Li-ion conductor, found to have excellent ionic conductivity of 2 mS cm⁻¹, was implemented in an all-solid-state battery set-up with promising interfacial stability⁶⁹. More recently, the Li_{0.388}Ta_{0.238}La_{0.475}Cl₃ superionic conductor, possessing excellent interfacial compatibility with lithium metal electrodes, has been reported^{70,71}. This material has large 1D channels that are ideal for rapid Li-ion transport and can be connected to form 3D migration networks using aliovalent doping at the La site. Although halide Na-ion conductors have also been reported, such as Na₃ErCl₆ and Na₂ZrCl₆ (refs. 72,73), their ionic conductivity is currently severely limited.

Materials modelling and machine learning methods

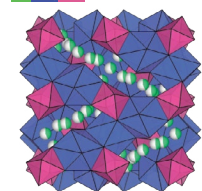
In general, two main classes of atomistic simulation techniques have been used to study solid-state electrolyte materials: electronic structure methods using density functional theory (DFT) and ab initio molecular dynamics (AIMD); and interatomic-potential-based (or forcefield) methods, including classical molecular dynamics (MD). More recently, there has been an increasing connection between these two classes through ML techniques in the context of materials understanding and discovery. Although not the focus of this Review, we note that other computational method developments include

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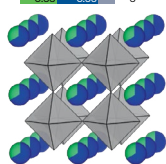
Lithium solid electrolytes

Oxides

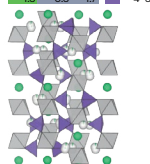
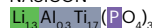
Garnet



Perovskite

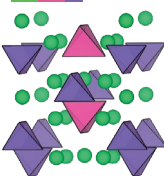


NASICON

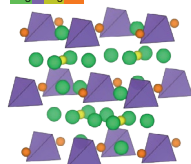


Sulfides

LGPS

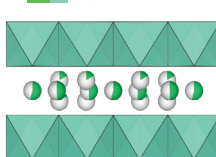


Argyrodite

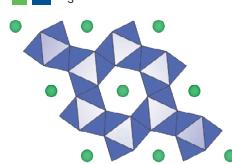


Halides

Monoclinic



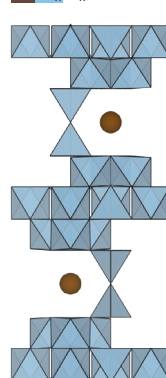
UCl3-type



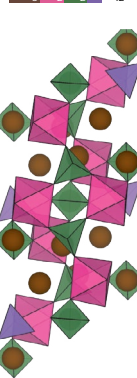
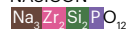
Sodium solid electrolytes

Oxides

β -alumina

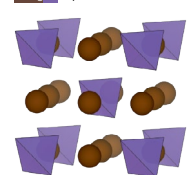


NASICON

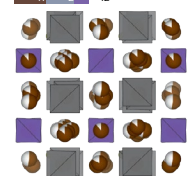


Sulfides

Thiophosphate

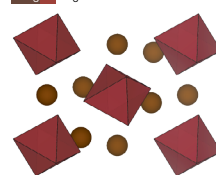


LGPS-like

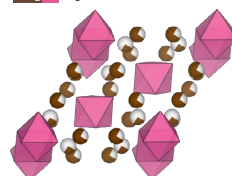


Halides

Monoclinic

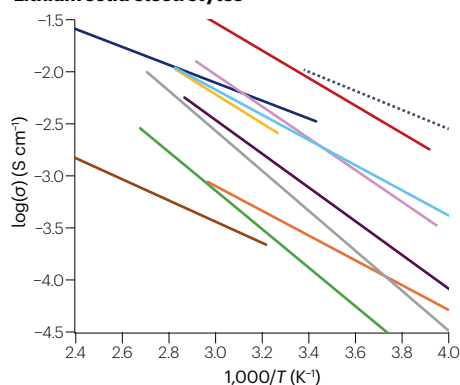


Trigonal



b

Lithium solid electrolytes



Sodium solid electrolytes

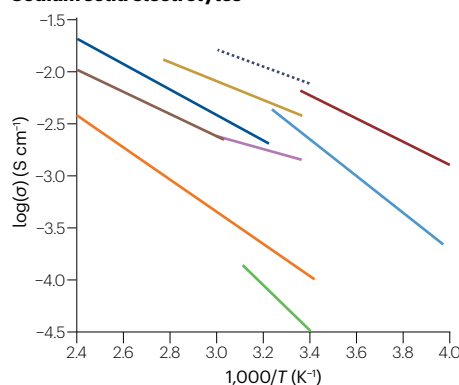


Fig. 2 | Crystal structures and ionic conductivities of topical solid electrolytes. a, Representative crystal structures of inorganic solid electrolyte materials for lithium and sodium solid-state batteries. **b,** Li-ion and Na-ion conductivities (σ) of inorganic crystalline solid electrolytes for solid-state batteries. Li-ion conductivities are taken from refs. 15,53,70,250,251. Na-ion conductivities are taken from refs. 60,72,252. Conductivities for commercial liquid electrolytes (LiPF_6 -ethylene carbonate (EC)/propylene carbonate (PC) and NaPF_6 -EC/dimethyl carbonate (DMC)) are also plotted for comparison. NASICON, Na superionic conductors.

the use of multiscale simulation tools capable of bridging the gap between different length scales from atomic scale to mesoscale and continuum modelling. Advances in materials modelling have been assisted by the rapid growth in computing power, which can be viewed as a pyramid where the base represents PCs and laptops, and at the apex are high-performance supercomputers now going beyond the terascale. An important aspect of solid-state battery and solid electrolyte research has been the strong synergy between computational modelling and experimental techniques (Fig. 3a). Here, we provide a brief overview of materials modelling and ML methods, as more detailed accounts of the theory behind these techniques are widely available^{74–77}.

Electronic structure methods

For solid-state battery materials, the most widely used computer modelling framework for the calculation of a system's electronic ground state is DFT. The Schrödinger equation for a many-body system may be simplified to the Kohn–Sham equation (a single-particle Schrödinger equation) and can be numerically solved with DFT⁷⁸. In the Kohn–Sham formalism, the total energy includes various terms: the electronic kinetic energy; the ‘Hartree term’, which describes the electrostatic energy resulting from the interaction of an electron with all other electrons and itself (that is, self-interaction); the energy attributed to

all nuclei; and the exchange–correlation energy, accounting for Pauli exclusion and electron correlation effects. For solid electrolytes, DFT simulations are regularly used to calculate, for example, the barriers for Li-ion and Na-ion hopping (via the nudged elastic band method), density of states, bandgaps, electronic conductivities, defect and polaron formation, stability and mechanical performance^{79–84}.

Electronic structure calculations can also be integrated with MD simulations to compute AIMD, which have proven to be valuable for investigating ion transport in a wide selection of solid electrolytes. Although computationally expensive and typically limited to <500 ions and timescales of hundreds of picoseconds, AIMD simulations can provide information on potential energy surfaces and activation energies for ionic conductivity⁸⁵.

Interatomic-potential-based (or forcefield) methods

Despite the substantial rise in the popularity of first-principles simulations, interatomic-potential-based (or forcefield) methods still have a critical role in materials modelling. The reliability and accuracy of these methods are highly dependent on the interatomic potentials used, which describe the interactions and total energy of a system. Potentials can be empirically derived using experimental and/or computational data^{86,87} or developed with ML of first-principles data, such as AIMD trajectories (as discussed below)^{88,89}.

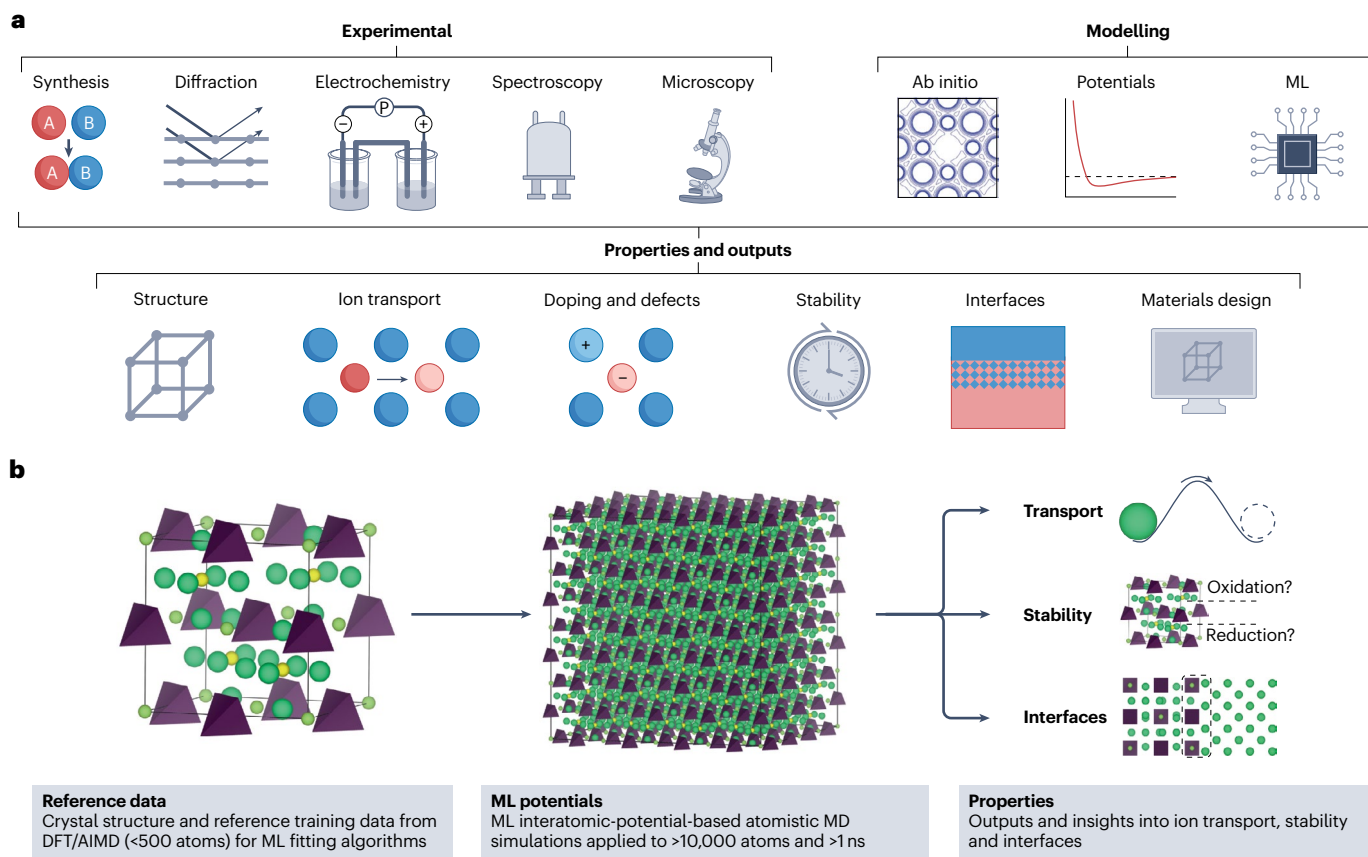


Fig. 3 | Methods and properties in the study of solid electrolytes and solid-state batteries. **a**, Schematic overview of the atomistic modelling and experimental techniques used to study solid electrolytes in solid-state batteries, as well as the most important properties and outputs that can be investigated with these techniques. **b**, Representative reference supercells for density

functional theory (DFT)/ab initio molecular dynamics (AIMD) datasets leading to large-scale machine learning (ML) interatomic-potential-based simulations of a solid electrolyte and the typical outputs (ion transport, stability trends and interfacial phenomena) obtained from such simulations. MD, molecular dynamics.

One of the primary applications of interatomic potentials is in MD simulations⁹⁰. In an MD simulation, Newton's equations of motion are solved using a numerical, iterative procedure over many time steps, leading to a clear image of the evolution of ion positions and velocities, which are updated using the velocities and forces, respectively, as a function of time. The chosen timestep for an MD simulation must be shorter than the typical time associated with important processes, such as an atomic vibration, with values of 1 or 2 fs often used. In the context of solid electrolytes for batteries, MD simulations are most often used to produce diffusion coefficients for mobile ions and to analyse the atomistic ion transport mechanisms as a function of temperature^{91,92}. Nevertheless, MD can also provide information on structural properties, for example radial distribution functions.

Substantial computing resources are required to run MD simulations that provide reliable statistics that can be directly compared with experimental results⁹¹. Fortunately, in the context of solid electrolytes, potential-based MD simulations can easily reach the nanosecond timescale with simulation boxes containing tens of thousands of atoms when implemented on modern supercomputing platforms.

Machine learning approaches

A rapidly growing addition to conventional materials modelling is the use of ML techniques^{12,75,93–105}. In this Review, we focus on the emerging approach of using ML-based interatomic potentials for MD simulations to probe increasingly complex issues of structural disorder, correlated conduction mechanisms and interfacial processes. The general approach in the development of accurate ML potentials usually involves using diverse and numerous *ab initio* datasets (typically generated from DFT calculations) to train the effective ML model. This training procedure then enables MD simulations based on ML potentials with orders of magnitude larger cell sizes and longer timescales than AIMD, while maintaining near quantum mechanical accuracy at reduced computational cost.

The general workflow follows three main steps (Fig. 3b): first, reference training data from high-level DFT or AIMD simulations and experimental structures (if available) are input for accurate fitting to derive ML potentials; second, ML potentials are validated and then applied for MD simulations that are much larger scale than current AIMD; and finally, these simulations generate valuable outputs on solid electrolyte properties, such as ion conduction mechanisms, stability trends and interfacial effects. It is worth stressing that the accuracy of these models depends upon the quality and diversity of the training data; for example, special care is needed to make sure that ion diffusion hops and pathways are appropriately sampled. Another challenge is the treatment of long-range electrostatic interactions by ML potentials, as describing ion migration in solid electrolytes and electrochemical systems requires simulating charged species, with possible changes in effective charge state along a diffusion path. This area is currently under method development with, for example, the implementation of long-range electrostatic interactions in neural network potentials over the past 2–3 years^{106,107}.

A brief overview of ML interatomic-potential development is provided here; more extensive reviews can be found elsewhere^{96,100}. The main types of ML regression algorithms that are currently used for deriving effective ML potentials include artificial neural networks (such as Behler–Parrinello networks¹⁰⁸) using sets of symmetry functions that are inspired by the network of neurons in the brain, with the training composed of modifying the weights between these functions or neurons; kernel-based regression methods¹⁰⁹, which encompass the Gaussian approximation potential framework, as well as automated

'on-the-fly' potentials coupled to AIMD; and linear fitting, which describes an atomic structure by appropriate basis functions and encompasses moment tensor potential approaches¹¹⁰.

In addition to ML potentials, the vast amounts of data that are readily generated through both experimental and computational approaches can be used as inputs for ML screening models. These models might be able to overcome some of the challenges, such as cost, scale and feasibility, associated with measurements and simulations using traditional methods, and they can also be used to identify design patterns that might otherwise be missed with typical analyses. A wide variety of ML approaches, including Bayesian networks, decision trees, artificial neural networks and support vector machines, has been used to predict and screen ionic conductivity in solid electrolytes^{93,111–116} and their synthesis and stability^{117–121}.

Understanding the mechanisms of fast-ion transport

Atomistic mechanisms, conduction pathways and activation energies that govern bulk Li-ion transport within solid electrolyte materials are important for understanding the rate capabilities of a solid-state battery. However, these mechanistic details are often difficult to extract on the atomic scale from experiment alone, especially correlated ion diffusion in complex compounds. In this section we show that materials modelling methods have been of considerable value in revealing details of the mechanisms and pathways of fast-ion conduction.

At the atomic level, the diffusion of mobile Li⁺ or Na⁺ cations within crystalline solids can be visualized as ion jumps or hops between ground-state sites and/or intermediate sites of the structural framework constituted of anions (such as O²⁻, S²⁻ or polyanion moieties)⁴. Cation vacancies or interstitials are typically the mobile charged species. The three principal migration mechanisms¹²² (Fig. 4a) are: vacancy diffusion, in which a neighbouring ion migrates into a vacant site; direct interstitial migration between adjacent interstitial sites; and a concerted or interstitialcy (knock-on) mechanism, where the migrating interstitial ion displaces a neighbouring lattice ion into the adjacent site. Vacancy and direct interstitial migration are seen as conventional hopping processes, whereas concerted migration is a highly correlated process in which fast-ion transport is affected by several ions moving together in a concerted manner and may include mechanisms involving correlated rotational motion.

We now outline the simulation studies that have elucidated atomic-scale mechanisms and dimensionality of conduction pathways in the most important classes of Li-ion and Na-ion crystalline inorganic solid electrolytes.

Garnets

Computational techniques have been used to investigate the transport mechanisms of garnet-type Li₇La₃Zr₂O₁₂ (LLZO)^{123–135}. AIMD simulations have revealed a concerted Li-ion conduction mechanism in cubic garnet LLZO¹²⁵, with most Li ions migrating in a highly concerted fashion¹³⁰ (Fig. 4b). Classical MD simulations¹²³ found that the Li distribution on tetrahedral and octahedral sites in cubic LLZO just above the phase transition temperature closely resembles the ordered distribution in tetragonal LLZO, and that the network of partially occupied sites strongly contributes to high conductivity with an activation energy (E_{act}) of 0.48 eV. An unsupervised landmark analysis¹³³ was used to detect Li-ion jumps in MD simulations of LLZO. Computational studies into doping LLZO indicated that incorporation of pentavalent cations (for example, Ta⁵⁺ and Nb⁵⁺) on the Zr sites stabilizes the cubic

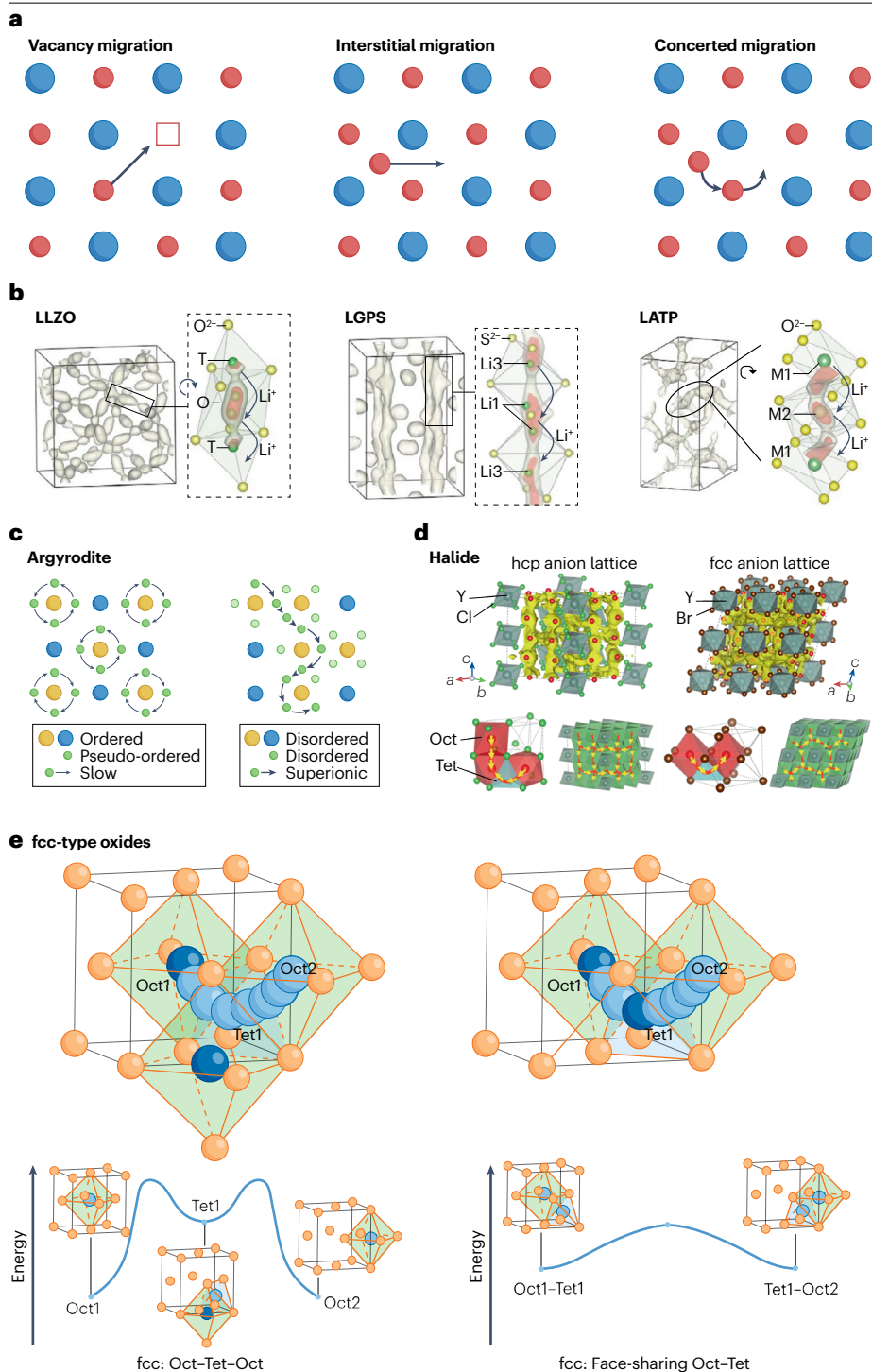


Fig. 4 | Modelling of ion migration mechanisms in solid electrolytes. **a**, Schematic representations of vacancy migration, direct interstitial migration, and concerted or interstitialcy migration, in which the migrating ion displaces a lattice ion into a neighbouring interstitial site. **b**, Probability density of Li^+ spatial occupancy in $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO, left), $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS, middle) and $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (LATP, right) during ab initio molecular dynamics (AIMD) simulations. The zoom-in subsets show the elongation feature of probability density along the migration channel (Li, green; O/S, yellow). The isosurfaces are $6\rho_0$, $6\rho_0$ and $2\rho_0$ for LGPS, LLZO and LATP, respectively, where ρ_0 is the mean probability density in each structure and the inner isosurfaces have twice the density of the outer isosurfaces. **c**, Effect of anion disorder on ion transport in argyrodites. **d**, Crystal structures (top) of Li_3YCl_6 with hexagonal close-packed (hcp)-like anion lattice and Li_3YBr_6 with face-centred cubic (fcc)-like anion lattice, superimposed with Li^+ probability density marked by yellow isosurfaces from AIMD simulations, and their corresponding Li-ion migration pathways (bottom). The red and blue polyhedra represent octahedral (Oct) and tetrahedral (Tet) interstitial sites, respectively. **e**, Oct-Tet-Oct pathway in fcc oxygen lattice without face-sharing Li in the initial state (top left) and its energy landscape (bottom left). Oct-Tet/Tet-Oct pathways in fcc oxygen lattice with face-sharing Li in the initial state (top right) and the energy landscape (bottom right). Orange, oxygen anions; dark blue, Li ions in the initial sites; light blue, Li ions in the migration paths. Panel **b** reprinted from ref. 130, CC BY 4.0. Panel **c** reprinted from ref. 162, CC BY 4.0. Panel **d** reprinted with permission from ref. 193, Wiley. Panel **e** reprinted from ref. 198, CC BY 4.0.

phase and enhances the ionic conductivity^{123,126,128}. Studies using data mining clustering techniques suggested that the presence of Ga^{3+} in octahedral sites, not tetrahedral sites, enhances Li-ion conductivity¹³⁵. Potential-based simulations of fluoride incorporation into LLZO compensated by lithium vacancies showed improved transport properties owing to stabilization of the cubic phase¹³¹.

$\text{Li}_{10}\text{GeP}_2\text{S}_{12}$

The Li-ion transport mechanisms in the superionic conductor $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ (LGPS) have been probed by numerous simulation studies^{136–143}, including the use of ML methods^{141,143}. Early MD studies^{136,137} predicted that, contrary to initial experimental speculation, LGPS has 1D diffusion channels along the *c* direction together with less facile diffusion pathways in the

ab plane (Fig. 4b); hence, LGPS can be viewed as a weakly anisotropic 3D Li-ion conductor. Nudged elastic band calculations revealed that the anion arrangement in this Li-ion conductor influences ion mobility¹³⁹: the body-centred cubic (bcc)-like sulfide framework of LGPS provides a direct pathway between tetrahedral sites, leading to the lowest activation barrier (0.15 eV) compared with other pathways in close-packed anion frameworks, such as hexagonal close-packed (hcp) or fcc. Neural network potentials were used to run up to 40-ns MD simulations that predicted a Li-ion sublattice phase transition at ~400 K, below which the *ab*-plane diffusivity is greatly reduced¹⁴³. To address a gap in understanding nanoscale effects in LGPS, nanoscale modelling techniques were used to predict that Li-ion conductivity increases with decreasing particle volume, owing to a change from a 1D to an isotropic 3D conduction mechanism and major local structural changes in the smallest nanoparticles¹⁴².

LISICONs

Among computational work on LISICON-related oxides^{144–148}, MD simulations¹⁴⁵ on the Li_4SiO_4 – Li_3PO_4 system revealed fast Li-ion conduction pathways through 3D networks in mixed Si/P compositions that follow a cooperative-type interstitial mechanism with E_{act} of 0.30–0.32 eV, consistent with the ion dynamics determined from NMR experiments. Modelling studies on the mixed polyanion effect in the LISICON oxide family $\text{Li}_{4+x}\text{Si}_{1-x}\text{X}_x\text{O}_4$ (X = P, Al or Ge)¹⁴⁶ revealed three temperature-dependent mechanisms for Li-ion diffusion: local oscillation at low temperature, isolated hopping at intermediate temperature, and superionic motion at high temperature, with the prediction that $\text{Li}_4\text{Al}_{1/3}\text{Si}_{5/6}\text{Ge}_{1/6}\text{P}_{1/3}\text{O}_4$ transitions to superionic motion at a temperature much lower than that of Li_4SiO_4 .

NASICONs

For Li-based NASICONs, nudged elastic band and AIMD calculations have been used to identify diffusion pathways, activation energies and cooperative transport mechanisms in $\text{Li}_{1.3}\text{Al}_{0.3}\text{Ti}_{1.7}(\text{PO}_4)_3$ (LATP)¹³⁰, $\text{Li}_{1+x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ (ref. 149) and $\text{Li}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ (ref. 150). For example, these methods revealed that Li-ion transport in LATP is dominated by concerted migrations of multiple ions with a low energy barrier of 0.27 eV (ref. 130) (Fig. 4b) and were used to investigate the influence of undercoordinated Li ions on the ionic conductivity of $\text{Li}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ (ref. 150).

Research into Na-based NASICONs has primarily focused on doping and polyanion mixing in $\text{Na}_{1+x}\text{Zr}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ (refs. 130, 150–153). MD simulations¹⁵³ conducted on $\text{Na}_3\text{Sc}_x\text{Zr}_{2-x}(\text{SiO}_4)_{2-x}(\text{PO}_4)_{1+x}$ and $\text{Na}_2\text{Sc}_y\text{Zr}_{2-y}(\text{SiO}_4)_{1-y}(\text{PO}_4)_{2+y}$ ($x, y = 0–1$) showed that Na-ion diffusion is 3D in these materials and that ionic conductivity decreases with a higher Sc^{3+} concentration. From >2,000 ab initio-based kinetic Monte Carlo simulations¹⁵² performed on $\text{Na}_{1+x}\text{Zr}_2\text{Si}_x\text{P}_{3-x}\text{O}_{12}$ ($0 \leq x \leq 3$) NASICONs to delineate their compositional space over broad time ranges, spatial resolutions and temperatures, it was found that both polyanion mixing and doping can greatly enhance Na-ion transport. A study using DFT simulations¹⁵⁴ considered both single ion hopping and concerted migration, defined as occupancy-conserved hopping, in several Na-based NASICONs. Concerted Na-ion migration was found to reduce the activation barrier only when Na ions are strongly ordered, for example in $\text{Na}_2\text{Sb}_2(\text{SiO}_4)_3$. By contrast, in $\text{Na}_3\text{Sc}_2(\text{PO}_4)_3$, where there is a tendency for minimal Na ordering, both single Na-ion hopping and concerted Na-ion migration exhibit similar energy barriers.

Argyrodites

The mechanisms of Li-ion conduction in the argyrodite family based on $\text{Li}_6\text{PS}_5\text{X}$ (where X = Cl, Br or I) have been investigated by numerous

modelling studies^{155–166}. Early MD and NMR work¹⁵⁵ found that a series of interstitial sites correspond to the centres of tetrahedra of S/X atoms with the connectivity between sites identified as the key restriction to ionic conductivity. DFT-based MD studies¹⁵⁶ suggested that besides Li-ion vacancies, the large difference in Li-ion conductivity between $\text{Li}_6\text{PS}_5\text{Cl}$ and $\text{Li}_6\text{PS}_5\text{I}$ is caused by the halide ion distribution over the available crystallographic sites. Combined experimental and AIMD work¹⁶⁰ showed that the site disorder affects the lithium substructure by decreasing the jump distance between ‘cages’ of clustered Li^+ ions. The expansion of these cages makes a more interconnected pathway for Li^+ diffusion, thereby increasing ionic conductivity with E_{act} of ~0.15 eV. To further probe the role of anion disorder, AIMD¹⁶² was used on $\text{Li}_6\text{PS}_5\text{X}$ (X = Cl or I) and identified three partially occupied Li sites that define a 3D network of face-sharing tetrahedra with active diffusion pathways (Fig. 4c) that give a percolating 3D diffusion network. The effect of local S/X coordination on Li-ion hopping activation energies was investigated¹⁶⁷, combined with percolation studies to show how a balanced distribution of S/X on the shared sites leads to optimal macroscopic conductivity. AIMD simulations of a BH_4 -substituted Li argyrodite suggested a weak correlation of Li diffusion with the B–H bond motion, with the enhanced conductivity mainly originating from the interaction between Li and BH_4 and no evidence of a paddlewheel effect from the polyanion¹⁶⁸.

β -aluminas and β'' -aluminas

β -alumina and β'' -alumina materials were among the first solid electrolytes to have their ion transport mechanisms investigated using atomistic modelling. Early studies used interatomic-potential-based defect and MD simulations to quantify and understand Na-ion motion in β -aluminas and β'' -aluminas^{169–174}. These studies revealed the highly correlated nature of the 2D Na-ion diffusion through the conduction plane containing only mobile cations and bridging oxygen ions in these materials. Interstitialcy diffusion was determined to be the dominant mechanism in β -alumina, whereas vacancy diffusion governs ion transport in β'' -alumina. More recently, these techniques were extrapolated to far greater scales to study anomalous diffusion¹⁷⁵ and mimic an experimental terahertz pump in frequency and magnitude through the inclusion of a pulsed electrical field¹⁷⁶ during simulations on Na-, Ag- and K- β - and β'' -aluminas. These studies showed that the distribution of charge-compensating defects, modulated by processing, drives static and dynamic disorder and leads to persistent subdiffusive ion transport at macroscopic timescales, and that the characterization of memory effects in transport connects atomistic defect chemistry to macroscopic performance. Hence, the findings from these two studies^{175,176} are important to enable the mechanism-driven ‘atoms-to-device’ optimization of fast-ion conductors.

Perovskites and antiperovskites

Atomistic modelling has been readily used to understand ion transport in perovskite solid electrolytes, primarily with the formula $\text{Li}_{3x}\text{La}_{2/3-x}\text{TiO}_3$ (LLTO) and its many doped variations^{177–181}. One of the earliest atomistic studies of LLTO used interatomic-potential-based MD to reveal that Li-ion diffusion occurs through the vacancy mechanism and to explore the correlations between size of bottleneck, diffusion coefficients and activation energies¹⁷⁷. Based on nudged elastic band calculations¹⁷⁸, LLTO as a cathode coating was found to have low activation barriers of 0.23–0.39 eV for Li-ion diffusion over a range of Li concentrations. Subsequently, several AIMD studies of Li-ion transport in undoped and doped LLTO have been carried out to quantify the mobility of Li ions

and identify their migration pathways as a function of Li concentration and temperature^{179–181}.

With respect to antiperovskite materials^{118,182–192}, DFT simulations¹⁸² identified a three-ion hop mechanism involving Li interstitial dumbbells in Li_3OCl , Li_3OBr and $\text{Li}_3\text{OCl}_{0.5}\text{Br}_{0.5}$. However, the high formation energies found for Li interstitials suggest that this mechanism is not prevalent in real samples. Later studies^{183,184} revealed similarly high energies for Li and Na interstitial formation, and a consensus has now been reached that cation vacancy migration (with E_{act} of ~ 0.25 – 0.31 eV) dominates in Li-based and Na-based antiperovskites¹¹⁸. The paddlewheel effect is another topic of interest for antiperovskite studies^{188–192}. For example, the relationship between Li-ion transport and rotation of the OH^- group in $\text{Li}_{3-x}\text{OH}_x\text{X}$ ($x = 0$ – 1 ; $\text{X} = \text{Cl}$ or Br) has been explored in detail using DFT and AIMD simulations^{188,189}, indicating that Li-ion conduction is highly correlated between the proton and Li-ion vacancy concentrations. In $\text{Na}_{3-x}\text{O}_{1-x}(\text{NH}_2)_x(\text{BH}_4)_x$, the existence of a double paddlewheel effect has been proposed, where the rotational mobility of both anion groups promotes fast Na-ion conduction¹⁹¹.

Halides

Li_3MX_6 ($\text{M} = \text{Y}$, In or Sc ; $\text{X} = \text{Cl}$ or Br) solid electrolytes can be grouped into fcc-based and hcp-based materials (Fig. 4d). AIMD simulations have revealed that in fcc-based structures, migration between octahedral sites takes place via a tetrahedral intermediate, whereas in hcp-based structures, low-barrier direct hopping between octahedral sites is also possible¹⁹³. These mechanisms lead to a largely anisotropic 3D diffusion network in fcc-based structures and interconnected 1D diffusion channels (with E_{act} of ~ 0.25 eV) in hcp-based structures. DFT simulations¹⁹⁴ suggested that to achieve optimized ion conduction, a low Li-ion content of ~ 40 – 60% octahedral occupancy and a sparse M^{3+} -ion distribution are necessary.

Design approaches for materials with fast-ion transport

In addition to improving fundamental understanding of transport mechanisms, a growing number of computational investigations^{56,73,117,130,139,195–201} have proposed design principles to help accelerate the discovery of new solid electrolyte materials. These studies largely identify structural features and compositions that promote fast-ion behaviour, which is typically a room-temperature conductivity of >1 mS cm^{-1} and an activation energy of <0.3 eV.

Design strategies for fast-ion transport have traditionally focused on minimizing cation coordination changes in transport pathways. By contrast, the packing of two anions, as in binary intermetallics, was investigated as a possible route to superionic Li-ion conductivity⁵⁶. This approach led to a new Li–Si–S–I-based conductor whose sulfide and iodide ordering combines aspects of hcp and fcc lattices, analogous to the structure of NiZr . High Li-ion conductivity was designed in fcc-type oxides, including a new Li–In–Sn–O-based compound, by creating face-sharing Li configurations through Li-excess over-stoichiometry in rocksalt-type structures¹⁹⁸ (Fig. 4e). Other work¹³⁹ indicated that the topology of the anion arrangement is the key factor in determining intrinsic Li-ion mobility and suggested that an underlying bcc-like anion framework, which allows direct Li hops between adjacent tetrahedral sites, is most desirable for achieving high ionic conductivity.

High-entropy cation mixing, which introduces local distortions into the material, was shown to improve ionic conductivity in Li–NASICON, Na–NASICON and Li–garnet structures¹⁹⁹. The distortions produce an overlapping distribution of site energies that promotes diffusion of alkali ions. The influence of the ‘paddlewheel’ effect, which

refers to a mechanism where the rotational motion of complex anions aids the mobility of Li and Na ions, has been reviewed²⁰¹. Strategies identified to develop fast-ion conductors that exploit the paddlewheel effect include the onset temperature of anion rotation being manipulated through entropy-driven mechanisms, or kinetically inhibited low-temperature trapping of high-temperature disorder. It was noted that comparable jump frequencies for anions and cations are necessary to achieve coupled anion rotation and cation diffusion.

To identify characteristic features of fast Li-ion conductors, the crystal structures of a wide selection of Li-conducting oxides and sulfides were examined²⁰⁰. The superionic conductors – many of which are based on polyanion groups of P, Si, Ge or B – were discovered to have enlarged Li sites, caused by local spaces in the structural framework, that promote fast Li-ion conduction. With respect to fast Na-ion conductors, whose diffusion mechanisms can differ from those of their Li-ion counterparts, a crystal structure analysis revealed the key presence of face-sharing high-coordination sites⁷³. Screening for this feature as a design principle led to the discovery of a family of Na-ion conductors $\text{Na}_x\text{M}_y\text{Cl}_6$ ($\text{M} = \text{La}$ – Sm) with the UCl_3 -type structure.

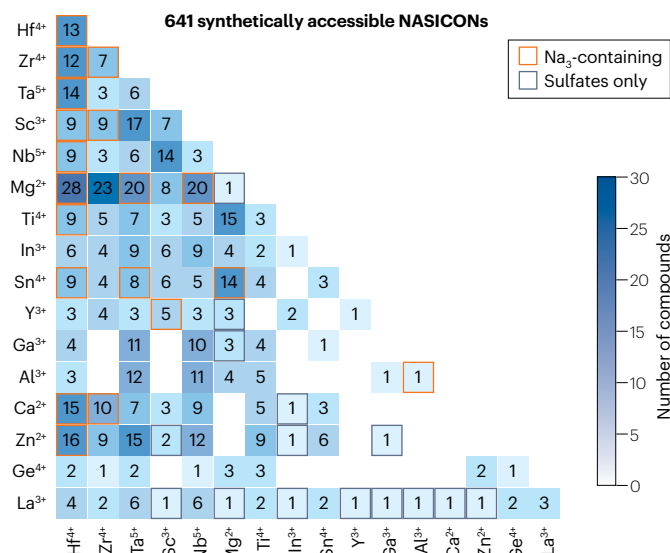
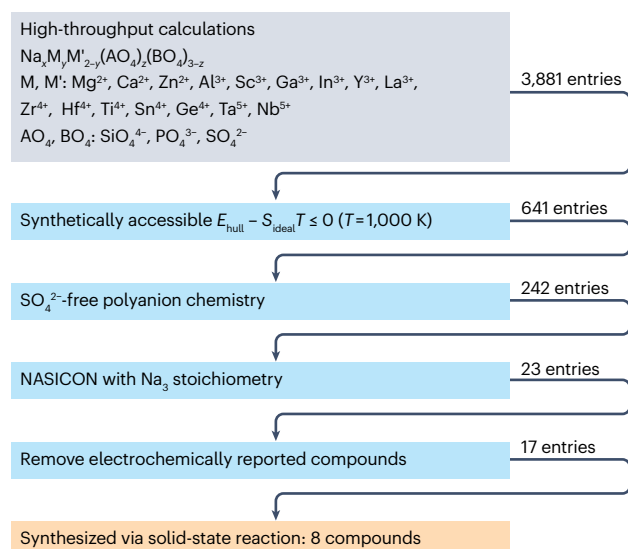
Predicting phase formation, stability and decomposition

In addition to featuring excellent ionic conductivity and interfacial compatibility, any solid electrolyte candidate must also be able to form a stable solid phase, especially for potential large-scale manufacturing. However, experimentally investigating these features can often be difficult and time consuming. Atomistic modelling can improve the time and resource efficiency of this task by delivering predictions regarding the likelihood of phase formation and stability that can guide future experiments in synthesis and device formulation (Fig. 5). In this section, we highlight key findings related to the phase formation, stability and decomposition mechanisms of promising solid electrolyte candidates.

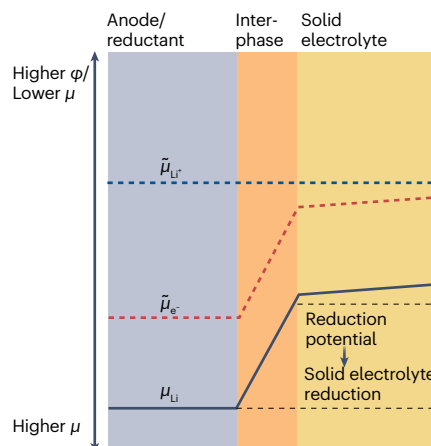
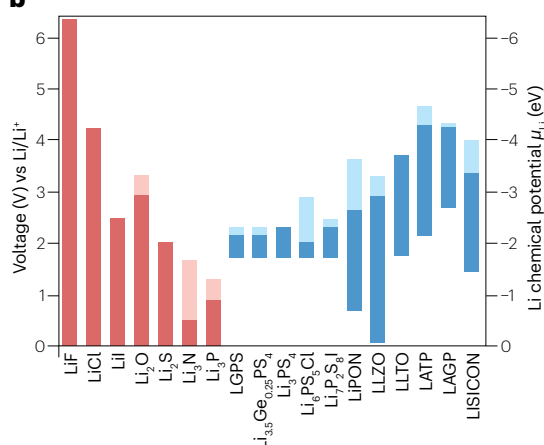
DFT calculations, including high-throughput methods, are important approaches for predicting the phase formation of solid electrolyte materials^{117,118,202,203}. Generally, solid electrolytes with substantial energies above the convex hull are considered unstable and synthetically inaccessible via typical routes^{41,79,197,198,202–205}. Combining computational predictions and experimental efforts provides a valuable synergy that can accelerate the development of synthetic approaches. New garnet-type solid electrolytes, including $\text{Li}_{6.5}\text{Nd}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$, $\text{Li}_{6.5}\text{Sm}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$ and $\text{Li}_{6.5}(\text{Sm}_{0.5}\text{La}_{0.5})_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$, were predicted by DFT and then successfully synthesized via an ultrafast sintering technique²⁰². More recently, eight NASICON-type solid electrolytes were realized from a first-principles synthesis map¹¹⁷ (Fig. 5a), one of which, $\text{Na}_{3.4}\text{Hf}_{0.6}\text{Sc}_{0.4}\text{ZrSi}_2\text{PO}_{12}$, achieved a room-temperature ionic conductivity of 1.2 mS cm^{-1} . A multistage AIMD sampling workflow and data-driven techniques²⁰³ revealed three promising (meta) stable sulfide solid solutions with high Na-ion conductivity, namely, $\text{Na}_{5-2x}\text{Al}_{1-x}\text{V}_x\text{S}_4$, $\text{Na}_{5-2x}\text{Al}_{1-x}\text{Ta}_x\text{S}_4$ and $\text{Na}_{5-2x}\text{In}_{1-x}\text{Sb}_x\text{S}_4$ ($0.375 \leq x \leq 0.625$). Atomistic simulations can also be used to elucidate how structural factors can affect the likelihood of solid electrolyte phase formation. A linear correlation between the stabilization temperature and the degree of lattice distortion in antiperovskite materials was observed via DFT simulations and the quasiharmonic approximation²⁰⁶, with higher synthesis temperatures required to realize candidates with higher ionic conductivities.

In the past 5 years, ML techniques have flourished in helping to assess the phase formation of solid electrolytes^{41,118,119}. The computational costs of screening a wide range of materials have made ML

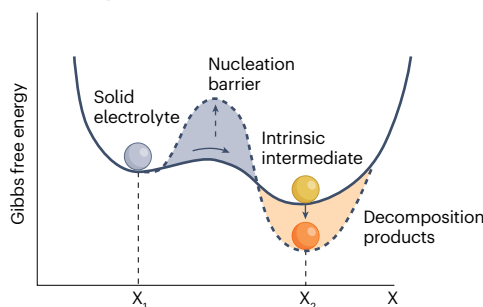
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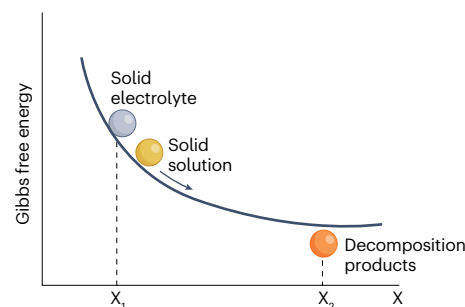
b



c Phase separation



Solid solution



methods highly complementary to DFT-based assessments. For example, a computational framework was developed to establish stability rules for NASICON-type solid electrolytes using ML-based phenomenological models⁴¹. Sure independence screening, a method based on correlation learning, and ML ranking algorithms were combined to search for the most suitable mathematical representation of NASICON stability. Within this computational paradigm, the synthetic

accessibility of systems at a finite temperature was estimated using an ideal mixing entropy model and the convex hull energies. The ab initio predictive capability was validated through the successful synthesis of five out of six predicted NASICONs.

One of the many challenges of assessing the phase formation and stability of a solid electrolyte is that such characteristics often contradict one another or impair other desirable features, such as high ionic

Fig. 5 | Modelling of phase formation, stability and decomposition in solid electrolytes. **a**, High-throughput computational workflow for identifying synthesizable Na superionic conductors (NASICONs) (left) and a distribution of 641 density functional theory (DFT)-calculated synthetically accessible NASICONs (right). The colour map indicates the number of stable compounds containing a specific metal pair. The squares outlined in orange indicate blocks including NASICONs with Na₃ stoichiometry and the squares outlined in grey mark blocks containing only sulfate-type NASICONs. **b**, Calculated electrochemical windows of solid electrolytes with the oxidation potential to fully delithiate the materials marked by the dashed line (left). Schematic showing

change of Li chemical potential μ_{Li} (black line), electrochemical potential $\tilde{\mu}_{\text{Li}}^+$ (blue dashed line) and $\tilde{\mu}_{\text{e}^-}$ (red dashed line) across the interface between the anode and the solid electrolyte (right). **c**, Schematic energy diagrams versus Li composition in a solid electrolyte for a first-order phase and solid solution reaction based on DFT-computed decomposition routes. LAGP, lithium aluminium germanium phosphate; LATP, lithium aluminium titanium phosphate; LGPS, Li₁₀GeP₂S₁₂; LISICON, Li superionic conductor; LLTO, Li_{0.5}La_{0.5}TiO₃; LLZO, Li₇La₃Zr₂O₁₂. Panel **a** reprinted from ref. 117, CC BY 4.0. Panel **b** adapted with permission from ref. 216, American Chemical Society. Panel **c** adapted from ref. 218, CC BY-NC-ND 4.0.

conductivity. Solid electrolytes displaying, concomitantly, good synthesizability scores and low thermodynamic stabilities, or high Li-ion diffusion and instabilities, have been reported^{118,119}. DFT calculations²⁰⁷ revealed a trade-off between ionic mobility and stability in antiperovskite electrolytes, with larger lattice distortions potentially improving ionic migration but lowering thermodynamical stability. In this context, multitargeted studies that investigate solid electrolyte features simultaneously (such as phase formation, electrochemical and thermodynamic stability, and ionic conductivity) have become essential. One reported multitarget multifidelity workflow²⁰⁸ systematically evaluates ionic and electronic conductivity and thermodynamic and electrochemical stability. A surrogate model was used to identify transition-state structures during ionic diffusion, overcoming the use of expensive nudged elastic band calculations. The workflow was successful upon application to a pool of antiperovskite systems, suggesting 14 systems that had already been identified as solid electrolyte candidates by previous experimental and computational studies. Another multitargeted methodology template was developed²⁰⁹ using an ensemble of high-throughput DFT calculations and ML-driven techniques to investigate the chemical space of >10,000 in silico tetragonal and cubic antiperovskite compounds.

Accurately measuring and predicting the electrochemical stability windows of solid electrolytes is also challenging. Early experimental results using semiblocking electrodes reported very wide electrochemical stability windows for many solid electrolytes (for example, 0–5 V for both LGPS and LLZO^{45,55,210–214}). Later studies, however, questioned the established stability of solid electrolytes as incoherences were found²¹⁵. In this context, atomistic simulations are invaluable in predicting electrochemical stability windows. Figure 5b shows the calculated electrochemical windows of topical solid electrolytes and a schematic showing the change of Li chemical potential and electrochemical potential across the interface between the anode and the solid electrolyte²¹⁶. In a pioneering DFT study²¹⁷, a methodology was developed to examine the thermodynamics of formation of solid electrolyte interfacial phases and therefore establish electrochemical windows. Thiophosphate electrolytes were predicted to have especially narrow electrochemical stability windows, and several known electrolytes were found to be inherently stable but reactive with electrodes, with the potential to form passivating but ionically conducting barrier layers. First-principles calculations and experimental results²¹⁴ together revealed that LGPS and LLZO have electrochemical stability windows of 1.71–2.14 V and 0.05–2.91 V, respectively, which are substantially narrower than those previously reported based on semiblocking electrodes.

For many solid electrolytes, predicting the electrochemical stability window by using the formation energies of the solid electrolyte decomposition products may underestimate observed values^{211,216–218}. As the kinetic barriers for decomposition reactions are not considered in thermodynamic investigations, elucidating the mechanisms related

to the decomposition of solid electrolytes is challenging. Nevertheless, important insights into the intrinsic relationship between the electrochemical stability window of solid electrolytes and their decomposition mechanisms have been discovered. DFT calculations, X-ray diffraction, solid-state NMR and electrochemical measurements²¹¹ were used to understand the relationship between the electrochemical stability window and redox reactions in argyrodite, garnet and NASICON solid electrolyte materials. This study revealed that the favourable decomposition pathway is indirect, via (de)lithiated states of these solid electrolytes towards their more stable decomposition products. Consequently, the electrochemical stability window is usually wider than that determined by solely considering the stability of the decomposition products. As a later extension of this work, a computational framework was developed²¹⁸ that predicts the electrochemical stability of solid electrolyte materials considering their redox activity. The predictions agreed well with previously reported experimental results, suggesting that the electrochemical stability windows are not always controlled by the decomposition products but instead often arise from the intrinsic stability of the solid electrolyte itself (Fig. 5c).

Understanding interfacial resistance and reactivity

Interfacial resistance and instability represent major bottlenecks to the commercialization of solid-state batteries. The simulation of solid electrolyte–electrode interfaces and the grain boundaries and surfaces of solid electrolytes is therefore increasingly important. However, compared with the more well-established modelling of bulk crystalline materials, the atomistic modelling of interfaces in battery (and other energy) materials is less developed. Simulations of interfaces require the preparation of a surface slab that is placed in direct contact with either a vacuum (surface) or a slab of the same material (grain boundary) or a different material (heterogeneous interface) (Fig. 6a). The development of such interfacial models is challenging owing to their scale and the need to consider chemical composition, stability, disorder and defects, and thermal and mechanical damage to ensure that they are representative of real interfaces. Nevertheless, such models can still offer a wealth of valuable information about the performance of solid electrolytes and solid-state batteries at the atomic scale, as well as the potential for discovery of unique behaviour not exhibited in bulk materials. Here we discuss the insights gained from current modelling work, with in-depth discussion of the methods used to simulate interfaces in battery materials available elsewhere^{219–222}.

Surfaces and morphologies

Atomistic modelling is critical to investigating solid electrolyte surfaces given the experimental difficulty in directly probing such interfaces with atomic specificity²²³. Modelling studies of oxide^{224–226}, sulfide^{227,228}

and halide²²⁹ solid electrolyte surfaces have typically focused on their ion transport, stability, decomposition and electronic structure. For example, DFT was used to investigate the surface structures and

particle morphology of LLZO under various physical and chemical conditions²²⁴. Strong evidence of Li segregation at LLZO surfaces at typical synthesis conditions was found, and practical strategies to

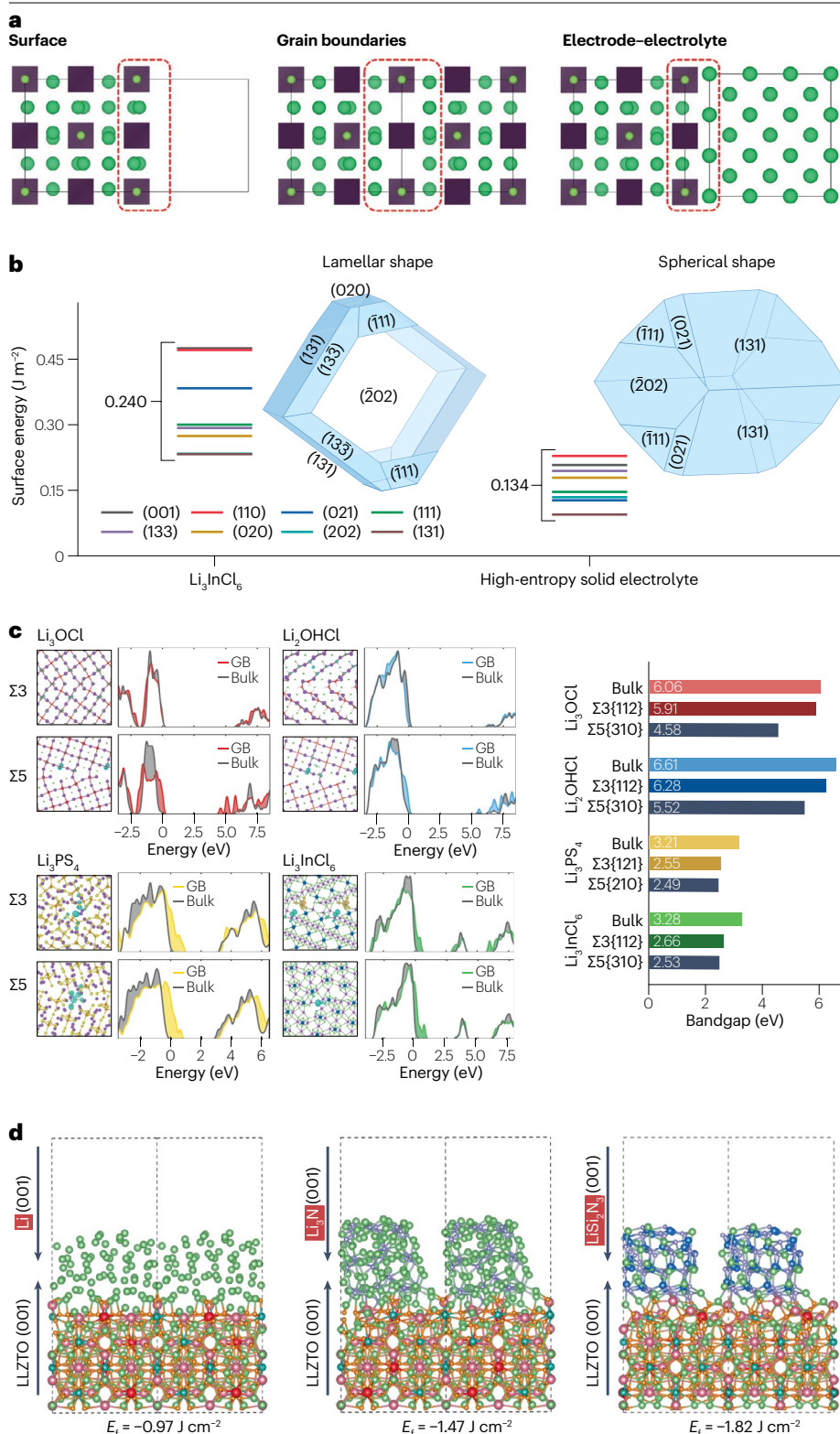


Fig. 6 | Modelling of interfaces in solid-state batteries. **a**, Schematics of the three main types of interfaces (surfaces, grain boundaries (GBs) and electrolyte–electrode interfaces) in solid electrolytes and solid-state batteries simulated using atomistic modelling. **b**, Calculated surface energies for typical crystal faces of Li₃InCl₆ and a high-entropy Li_{2.8}In_{0.2}Sc_{0.2}Yb_{0.2}Lu_{0.2}Zr_{0.2}Cl₆ solid electrolyte. Insets show Wulff shapes of the solid electrolytes. **c**, Projected density of states of selected grain boundaries in Li₃OCl, Li₂OHCl, Li₃PS₄ and Li₃InCl₆ and their corresponding bandgaps. **d**, Density functional theory calculations of the interfacial formation energies (E_f) of Li|Ta-doped Li₃La₂Zr₂O₁₂ (LLZTO), Li₃N|LLZTO and LiSi₂N₃|LLZTO interfaces. Panel **b** reprinted from ref. 229, CC BY 4.0. Panel **c** reprinted from ref. 196, CC BY 4.0. Panel **d** reprinted with permission from ref. 239, Wiley.

curb this were proposed. In another DFT study of LLZO surfaces²²⁵, substantial Li nucleation and electron trapping were predicted, thereby providing a potential route to Li dendrite formation. For argyrodites, DFT simulations were used to study the mechanical and electronic properties of the stable and metastable surfaces of $\text{Li}_6\text{PS}_5\text{Cl}$ and predict the morphology of its crystallites²²⁷. First-principles simulations can also be used to investigate decomposition at solid electrolyte surfaces; in one such example²²⁸, the reaction between LGPS surface and H_2O in the atmosphere was explored, finding that H_2O preferentially reacts with the sulfur atoms of the PS_4 tetrahedra to produce H_2S . The atomistic modelling of halide solid electrolyte surfaces is less developed than that of oxides and sulfides. Nevertheless, DFT was used to compute the surface energies of several distinct crystal facets of Li_3InCl_6 and a Li_3InCl_6 -based high-entropy halide, $\text{Li}_{2.8}\text{In}_{0.2}\text{Sc}_{0.2}\text{Yb}_{0.2}\text{Lu}_{0.2}\text{Zr}_{0.2}\text{Cl}_6$ (ref. 229) (Fig. 6b). These surface energies were then used to construct Wulff shapes for each material, which can be used to predict the preferential growth of crystal facets and hence the shape of crystalline particles.

Grain boundaries

Inorganic solid electrolytes are typically polycrystalline and therefore contain high densities of grain boundaries, which can present considerably different ionic conductivities, electronic structures and mechanical properties compared with the bulk material²²⁰. Given the detrimental impact that grain boundaries are known to have on ion transport in oxide solid electrolytes, it is unsurprising that modelling studies have focused on this area. Most of these studies feature antiperovskite, perovskite and garnet solid electrolytes. For example, in the first studies to build atomistic models of grain boundaries for battery solid electrolytes, classical MD simulations were used to investigate Li-ion transport in large-scale $\Sigma 3$ and $\Sigma 5$ GBs in Li_3OCl (ref. 230) and LLZO²⁹. Both studies found that Li-ion transport was hindered at the grain boundaries, with activation energies for Li-ion conduction across the grain boundaries consistently higher than those of the bulk systems, thereby demonstrating the high grain-boundary resistance in these oxide materials. Subsequent DFT and AIMD studies have also reported similar findings for hydrated antiperovskites (Li_2OHBr (ref. 231) and Li_2OHCl (ref. 196)) and Al-doped and Nb-doped LLZO²³². Hybrid Monte Carlo/MD simulations based on ML potentials²³³ were used to investigate why grain boundaries in the perovskite $\text{Li}_{0.375}\text{Sr}_{0.4375}\text{Ta}_{0.75}\text{Zr}_{0.25}\text{O}_3$ do not substantially reduce Li-ion transport in contrast to LLZO²³⁴. In contrast to oxides, negligible grain-boundary resistance has been confirmed by several MD and DFT studies of individual grain boundaries and polycrystals in a selection of sulfide and halide solid electrolytes, including Li_3PS_4 (ref. 196), Na_3PS_4 (ref. 235) and Li_3InCl_6 (refs. 66,196).

Atomistic modelling can also be used to ascertain the mechanical performance and electronic structure of grain boundaries and predict their influence on dendrite formation and propagation through solid electrolytes. For example, classical MD calculations performed on both tilt and twist grain boundaries in LLZO²³⁶ found that the grain boundaries were considerably softer than the bulk material, which correlates with observations of grain boundaries acting as sites for dendrite propagation²³⁷. A DFT study¹⁹⁶ of Li_3OCl , Li_2OHCl , Li_3PS_4 and Li_3InCl_6 reported reduced bandgaps in the vicinity of grain boundaries (Fig. 6c) and illustrated the role that polaron diffusion can play in the leakage current at grain boundaries in solid electrolytes.

Electrolyte–electrode interfacial reactions

Solid electrolytes can readily undergo a variety of interfacial reactions when in contact with electrode materials, which are often detrimental

to the performance of solid-state batteries because they can cause high interfacial resistance, low Coulombic efficiency and failure during cycling⁷⁹. Atomistic modelling and ML techniques have been used to study interfacial reactions because of the challenges associated with the experimental characterization of these hidden or buried interfaces.

For the anode–electrolyte interface, LLZO is one of the most well-studied solid electrolytes in combination with Li metal anodes. Using ultrathin Al_2O_3 as a coating to overcome the large interfacial resistance between LLZO and Li metal, explicit Li-metal– Al_2O_3 and Li-metal– Li_2CO_3 (present at garnet surface) interfacial models were constructed²³⁸. DFT and AIMD simulations of these models suggested that the oxide coating enables wetting of metallic Li and allows for effective Li-ion transport across the interface. Similar techniques have also been used to investigate interfacial transport and stability between LLZO and Li_3N or LiSi_2N_3 as additives to Li metal anodes²³⁹ (Fig. 6d). AIMD calculations²⁴⁰ found that the Li– $\text{Li}_6\text{PS}_5\text{Cl}$ interface decomposes with the formation of numerous decomposition products, including Li_2S , Li_3P and LiCl . A more recent study on the same system²⁴¹ presented atomistic simulations of solid electrolyte interphase growth with >30,000 atoms for 10 ns based on ab initio-trained ML interatomic potentials. A two-step growth mechanism was revealed, in which an amorphous argyrodite phase is formed, followed by crystallization of the reaction products into a $5\text{Li}_2\text{S}\cdot\text{Li}_3\text{P}\cdot\text{LiCl}$ solid solution. In another study, DFT calculations explained why LiF-rich solid electrolyte interphases can effectively block lithium dendrites based on their electronic structures²⁴². First-principles calculations²⁴³ were used to investigate the stability of >20 interfaces between the decomposition products of important solid electrolytes, such as Li_3PS_4 , $\text{Li}_6\text{PS}_5\text{Cl}$ and Na_3PS_4 , against Li and Na metal anodes, with the work of adhesion required to form such interfaces predicted to be critical in quantifying interfacial stability.

Several DFT studies have examined cathode–electrolyte interfaces^{244–246}. One such example elucidated the characteristics of space-charge layers at interfaces between oxide cathodes and sulfide electrolytes²⁴⁴. Another investigated the interfacial binding between surface facets of the LiCoO_2 cathode and the Li_3YCl_6 solid electrolyte, observing substantially different binding behaviour depending on the chemical composition of the Li_3YCl_6 surface termination²⁴⁷. These findings were correlated with the experimentally determined electrochemical stability and decomposition of this interface. The interfacial stability and reaction mechanisms of another promising halide solid electrolyte, $\text{Li}_3\text{YCl}_6\text{Br}$, with high-voltage $\text{Li}(\text{Ni},\text{Mn},\text{Co})\text{O}_2$ (NMC) cathodes and Li metal were investigated using AIMD simulations²⁴⁸. Although reactions were predicted between the solid electrolyte and cathode, with changes in the oxidation states of Br^- anions in $\text{Li}_3\text{YCl}_6\text{Br}$ and d -element cations in NMC, they were restricted to the interface. By contrast, the $\text{Li}_3\text{YCl}_6\text{Br}$ –Li interface exhibited substantial instability, with the formation of LiCl and LiBr at the solid electrolyte surface and metallic Y at the Li anode surface. Atomistic models of interfaces have also been considered in the context of solid-state Li–S batteries. Using a combination of DFT and MD calculations²⁴⁹, the thermodynamics and kinetics of typical cathode–electrolyte interfaces in solid-state Li–S batteries were explored. It was predicted that sulfides are more stable than oxides, nitrides and halides against the S_8 cathode. Large-scale models of thousands of atoms or more were also used to explore Li-ion diffusion at the S_8 – Li_3PS_4 interface.

Future perspectives

This Review has highlighted the exciting progress being made in understanding inorganic solid-state battery electrolytes using advanced atomistic modelling and the growing application of ML techniques.

Future computational studies are likely to address key challenges of solid electrolyte materials that are critical to solid-state battery development and will encompass the following themes.

Interfaces and stability

A deeper understanding of interfacial reactions and compatibilities between solid electrolytes and electrode materials, including metal anodes and high-voltage cathodes, is needed. Important aspects to focus on include resistance at the electrolyte–electrode interface and its effect on ionic transport during charge or discharge, as well as grain-boundary resistance, which can be a particular issue with ceramic oxide electrolytes. Further advances in thermodynamic models for electrochemical stability and interfacial reactivity are important to help guide intrinsic voltage limits for practical operation of solid electrolytes.

Design and optimization of new materials

High-throughput data-driven computational tools together with experimental input will continue to be used to investigate compositional changes and chemical doping in conventional solid electrolytes, and to help screen design factors for new fast-ion conductors. The numerous investigations in this area suggest that there is no single unified design principle that facilitates superionic behaviour, and that diverse design factors can be used to optimize a particular family of solid electrolyte material.

Machine learning and technique development

To enable substantially larger system sizes than standard *ab initio* techniques, the application of ML methods to probe ion transport and interfacial processes must be further improved. Because the accuracy of ML studies depends upon the quality of the training data, special care is required to make sure that there is diverse and appropriate sampling in the training configurations, and that there is rigorous validation of the accuracy of the fitting; indeed, without appropriate training data, the old adage ‘garbage in, garbage out’ applies. Other important technique developments include increased coupling between computation and experimental techniques (for example, NMR, nonlinear optics and neutron scattering), and the use of multiscale simulation tools capable of integrating different length scales from atomistic to continuum modelling.

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Author contributions

M.S.I. and J.A.D. conceived the theme and concept of the Review. All authors researched data, contributed substantially to the discussion of the content, wrote the manuscript and edited it before submission.

Competing interests

The authors declare no competing interests.

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